



Reading: Data formats in practice

When you think about the word "format," a lot of things might come to mind. Think of an advertisement for your favorite store. You might find it in the form of a print ad, a billboard, or even a commercial. The information is presented in the format that works best for you to take it in. The format of a dataset is a lot like that, and choosing the right format will help you manage and use your data in the best way possible.

Data format examples

As with most things, it is easier for definitions to click when you can pair them with examples you might encounter on a daily basis. Review each data format's definition first and then use the examples to lock in your understanding.

Primary versus secondary data

The following table highlights the differences between primary and secondary data and presents examples of each.

Data format classification	Definition	Examples
Primary data	Collected by a researcher from first-hand sources	• Data from an interview you conducted - Data from a survey returned from 20 participants • Data from questionnaires you got back from a group of workers
Secondary data	Gathered by other people or from other research	• Data you bought from a local data analytics firm's customer profiles • Demographic data collected by a university • Census data gathered by the federal government

Internal versus external data

The following table highlights the differences between internal and external data and presents examples of each.

Data format classification	Definition	Examples
Internal data	Data that is stored inside a company's own systems	- Wages of employees across different business units tracked by HR - Sales data by store location - Product inventory levels across distribution centers
External data	Data that is stored outside of a company or organization	- National average wages for the various positions throughout your organization - Credit reports for customers of an auto dealership

Continuous versus discrete data

The following table highlights the differences between continuous and discrete data and presents examples of each.

Data format classification	Definition	Examples
Continuous data	Data that is measured and can have almost any numeric value	- Height of kids in third grade classes (52.5 inches, 65.7 inches) - Runtime markers in a video - Temperature
Discrete data	Data that is counted and has a limited number of values	- Number of people who visit a hospital on a daily basis (10, 20, 200) - Maximum capacity allowed in a room - Tickets sold in the current month

Qualitative versus quantitative data

The following table highlights the differences between qualitative and quantitative data and presents examples of each.

Data format classification	Definition	Examples
Qualitative	A subjective and explanatory measure of a quality or characteristic	• Favorite exercise activity • Brand with best customer service • Fashion preferences of young adults
Quantitative	A specific and objective measure, such as a number, quantity, or range	• Percentage of board certified doctors who are women • Population size of elephants in Africa • Distance from Earth to Mars at a particular time

Nominal versus ordinal data

The following table highlights the differences between nominal and ordinal data and presents examples of each.

Data format classification	Definition	Examples
Nominal	A type of qualitative data that is categorized without a set order	• First time customer, returning customer, regular customer • New job applicant, existing applicant, internal applicant • New listing, reduced price listing, foreclosure
Ordinal	A type of qualitative data with a set order or scale	• Movie ratings (number of stars: 1 star, 2 stars, 3 stars) • Ranked-choice voting selections (1st, 2nd, 3rd) • Satisfaction level measured in a survey (satisfied, neutral, dissatisfied)

Structured versus unstructured data

The following table highlights the differences between structured and unstructured data and presents examples of each.

Data format classification	Definition	Examples
Structured data	Data organized in a certain format, like rows and columns	• Expense reports • Tax returns • Store inventory
Unstructured data	Data that cannot be stored as columns and rows in a relational database.	• Social media posts • Emails • Videos
